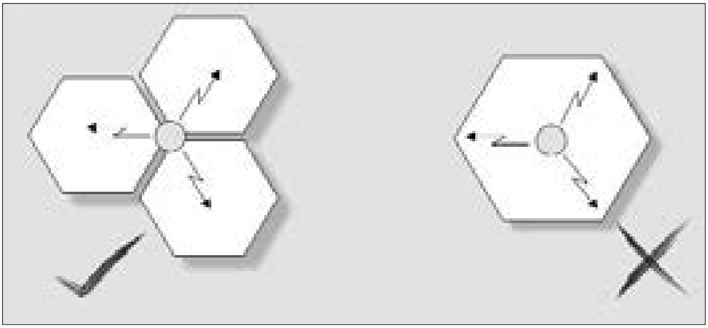

Busting the misconception - cellular antennas are actually located on the corners of cells, not at their center

Another thing that resulted from the independence of these cells was the concept of *frequency reuse*. To illustrate this, suppose the government allowed operator A to use the 800 MHz - 900 MHz range of frequencies for his cellular operation. Let's say A calculated that 400 different channels would be possible in this range of frequencies. Now because each cell can be treated independently, each base station could support the 800 - 900 MHz range with 400 channels. Following this, it could mean there could be 400 calls made *from each cell* at any given time. This system, however, would not work; signals from neighbouring cells would interfere with each other, resulting in nonsensical data cluttering the airwaves. The network had to be designed so that no cell would be next to any other cell that used the same set of frequencies.

Next on the agenda was deciding the shape of the cell. It had to be made such that frequency bands would be usable as much as possible. This meant that the number of cells surrounding a central cell would have to be the minimum possible. Would it be circular, triangular, square... what? After considerable thought, the designers of the system realised that Nature had already solved that problem: the answer was the Hexagon.

Hexagons fit neatly into each other, and each hexagon is surrounded by only six others. Comparing this with the square (eight surrounding squares) or the triangle (twelve surrounding trian-